

Introduction to Basic and Advanced Statistical Modelling

Using R (solutions to exercises)

Alexandre Courtiol

Table of contents

Reading data

Wrangling data

Wrangling spatial data

Plotting data

Reading data

Practice 4

```
territories <- read_csv("data_MHB/data-territories.csv")
```

```
Rows: 1152306 Columns: 10
```

```
-- Column specification -----
```

```
Delimiter: ","
```

```
chr (2): nameeg, namelt
```

```
dbl (8): species_id, euring_id, samplearea_id, year, territories_visit_1, territories_...
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

Wrangling data

Practice 5a

Use `select()` and/or `filter()` and/or `slice()` and/or `arrange()` to determine:

1. the largest number of recorded species across all years

```
species_count |>
  arrange(desc(n_recorded_species)) |>
  slice(1) |>
  select(n_recorded_species)
```

or

```
species_count |>
  filter(n_recorded_species == max(n_recorded_species)) |>
  select(n_recorded_species)
```

Practice 5b

Use `select()`, `filter()`, `slice()` and/or `arrange()` to determine:

2. the largest number of recorded species in 2025

```
species_count |>
  filter(year == 2025) |>
  filter(n_recorded_species == max(n_recorded_species)) |>
  select(n_recorded_species)
```

but NOT this:

```
species_count |>
  filter(year == 2025, n_recorded_species == max(n_recorded_species)) |>
  select(n_recorded_species)
```

Do you understand why?

Practice 6a

Use the functions we have seen to obtain:

1. the range of median elevation according to the number of visits

```
species_count |>
  summarise(median_elevation_min = min(median_elevation),
            median_elevation_max = max(median_elevation),
            .by = n_visits)
```

```
# A tibble: 2 x 3
```

	n_visits	median_elevation_min	median_elevation_max
	<dbl>	<dbl>	<dbl>
1	3	250	2150
2	2	1450	2750

Practice 6b

Use the functions we have seen to obtain:

2. the 3 most recorded species for each year

```
territories |>
  summarise(territories_sum = sum(territories_total),
            .by = c(year, nameeg, namelt)) |> # aggregate across samplearea_id
  slice_max(territories_sum, n = 3, by = year) |> # pick top 3 per year
  select(year, nameeg, namelt, territories_sum)
```

A tibble: 81 x 4

	year	nameeg	namelt	territories_sum
	<dbl>	<chr>	<chr>	<dbl>
1	1999	Eurasian Chaffinch	Fringilla coelebs	5775
2	1999	Common Blackbird	Turdus merula	3132
3	1999	Coal Tit	Periparus ater	3108
4	2000	Eurasian Chaffinch	Fringilla coelebs	6695

i 77 more rows

Practice 6c (1)

Use the functions we have seen to obtain:

3. the number of observations for *Sturnus vulgaris* for each year

```
territories |>
  filter(namelt == "Sturnus vulgaris") |>
  summarise(territories_sum = sum(territories_total, na.rm = TRUE), # NAs present
            .by = year)
```

```
# A tibble: 27 x 2
```

	year	territories_sum
	<dbl>	<dbl>
1	1999	691
2	2000	920
3	2001	911
4	2002	887

```
# i 23 more rows
```

Practice 6c (2)

Use the functions we have seen to obtain:

3. the number of observations for *Sturnus vulgaris* for each year

```
territories |>
  filter(namelt == "Sturnus vulgaris") |>
  drop_na(territories_total) |> # alternative not to use na.rm below
  summarise(territories_sum = sum(territories_total), # NAs present
            .by = year)
```

```
# A tibble: 27 x 2
  year territories_sum
<dbl>           <dbl>
1  1999             691
2  2000             920
3  2001             911
4  2002             887
# i 23 more rows
```

Practice 7

Try to use `pivot_longer()` to get back to the original long format of the data from the output of the following code:

```
territories |>
  filter(year == 2025) |>
  select(samplearea_id, nameeg, territories_total) |>
  pivot_wider(names_from = nameeg,
              values_from = territories_total) |>
  pivot_longer(-samplearea_id,
              names_to = "nameeg",
              values_to = "territories_total")
```

A tibble: 42,282 x 3

	samplearea_id	nameeg	territories_total
	<dbl>	<chr>	<dbl>
1	268	Little Grebe	0
2	268	Great Crested Grebe	0
3	268	Grey Heron	0
4	268	Little Bittern	0

i 42,278 more rows

Wrangling spatial data

Practice 8a

Use the functions we have seen to obtain:

1. the sampled area (sensu convex hull) across years

```
species_count_sf |>
  summarise(geometry = st_combine(geometry), .by = year) |>
  mutate(hull = st_convex_hull(geometry),
         area = st_area(hull),
         area_km2 = units::set_units(area, "km^2")) |>
  select(-hull) |>
  st_drop_geometry()
```

```
# A tibble: 27 x 3
```

	year	area	area_km2
* <dbl>		[m^2]	[km^2]
1	1999	48899915383.	48900.
2	2000	48995663682.	48996.
3	2001	48229824468.	48230.
4	2002	48995663682.	48996.

```
# i 23 more rows
```

Practice 8b (1)

Use the functions we have seen to obtain:

2. the IDs of the sampled areas which are most peripheral

Simple sorting by distance to centroid (but lacks threshold)

```
species_count_sf |>
  distinct(samplearea_id, geometry) |>
  mutate(centroid = st_centroid(st_combine(geometry)),
         dist = st_distance(centroid, geometry, by_element = TRUE)) |>
  arrange(desc(dist)) |>
  as_tibble() # (for slide only -> skip metadata clutter)
```

A tibble: 267 x 4

	samplearea_id	geometry	centroid	dist
	<dbl>	<POINT [°]>	<POINT [°]>	[m]
1	269	(6.110458 46.21013)	(8.234991 46.79076)	174955.
2	398	(10.40901 46.61099)	(8.234991 46.79076)	166982.
3	271	(6.184874 46.35493)	(8.234991 46.79076)	164026.
4	397	(10.34234 46.82876)	(8.234991 46.79076)	160429.

i 263 more rows

Practice 8b (2)

More useful, but more complex: intersection with concave hull outline

```
species_count_sf |>
  distinct(samplearea_id, .keep_all = TRUE) |>
  mutate(all_points = st_combine(geometry), # combine points into new geometry
         hull = st_concave_hull(all_points, ratio = 0.2), # create concave hull
         perimeter = st_boundary(hull), # extract outer line of the hull
         perimeter = st_buffer(perimeter, dist = 1000), # create buffer zone around it
         on_perim = st_intersects(perimeter, geometry, # test if points fall into buffer
                                   by_element = TRUE)) -> data_perimeter_sf

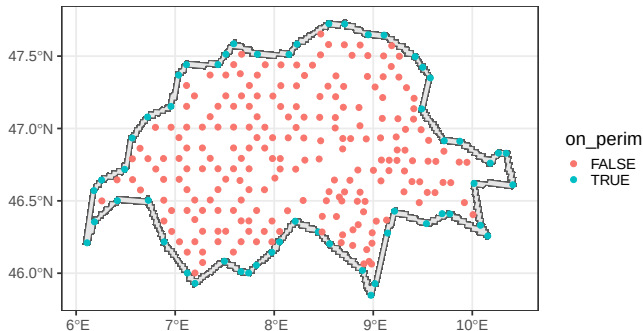
data_perimeter_sf |>
  filter(on_perim) |>
  pull(samplearea_id) # pull() extracts a column
```

```
[1] 269 271 273 277 278 279 282 283 286 289 291 294 302 307 308 311 323 325 340 352 359
[22] 369 370 376 384 385 388 392 393 394 395 396 397 398 400 408 411 418 427 434 439 456
[43] 460 461 462 469 475 476 478 502 503 515 522 525 531
```


Practice 8b (2)

More useful, but more complex: intersection with concave hull outline

```
theme_set(theme_bw(base_size = 20)) # setting the theme for all plots
ggplot(data_perimeter_sf) +
  geom_sf(aes(geometry = perimeter)) +
  geom_sf(aes(colour = on_perim)) # geometry = geometry is implicit
```



Plotting data

Practice 10 (1)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

```
territories |>
  summarise(territories_sum = sum(territories_total, na.rm = TRUE),
            .by = c(nameeg, namelt)) |>
  slice_max(territories_sum, n = 6) |>
  pull(namelt) -> top6sp
```

top6sp

```
[1] "Fringilla coelebs" "Sylvia atricapilla" "Turdus merula"      "Periparus ater"
[5] "Erithacus rubecula" "Passer domesticus"
```

Practice 10 (2)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

```
territories |>
  filter(namelt %in% top6sp) |>
  summarise(territories_sum = sum(territories_total, na.rm = TRUE),
            .by = c(nameeg, namelt, year)) -> top6_across_time
```

```
top6_across_time
```

```
# A tibble: 162 x 4
```

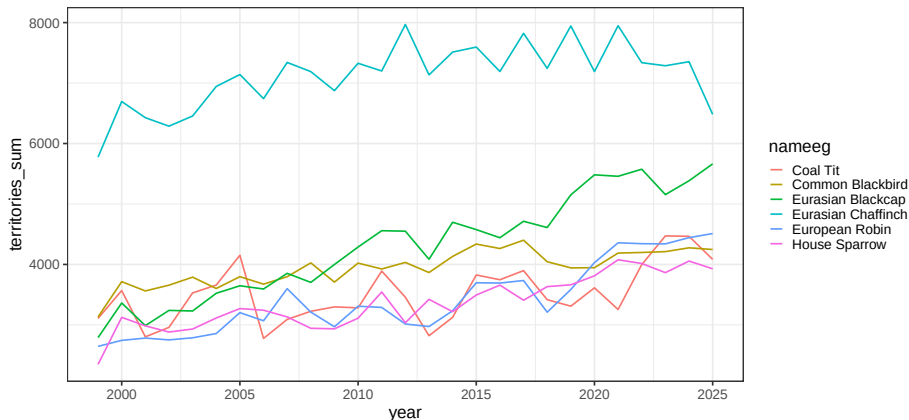
	nameeg	namelt	year	territories_sum
	<chr>	<chr>	<dbl>	<dbl>
1	Coal Tit	Periparus ater	1999	3108
2	Coal Tit	Periparus ater	2000	3568
3	Coal Tit	Periparus ater	2001	2803
4	Coal Tit	Periparus ater	2002	2959

```
# i 158 more rows
```

Practice 10 (3a)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

```
ggplot(top6_across_time) +  
  geom_line(aes(y = territories_sum, x = year, colour = nameeg))
```



Practice 10 (3b)

- combine data wrangling and plotting to illustrate the change in species abundance across years for the top 6 most encountered species

```
ggplot(top6_across_time) +  
  geom_line(aes(y = territories_sum, x = year)) +  
  facet_wrap(~ nameeg, scales = "free_y")
```

